

Received 17 November 2024, accepted 30 December 2024, date of publication 3 January 2025, date of current version 8 January 2025.

Digital Object Identifier 10.1109/ACCESS.2024.3525459

APPLIED RESEARCH

Machine and Deep Learning Models for Stress Detection Using Multimodal Physiological Data

EMAN ABDELFAH¹, SHREEHAR JOSHI², AND SHREEKAR TIWARI³

¹School of Computer Science and Engineering, Sacred Heart University, Fairfield, CT 06825, USA

²The Boring Company, Las Vegas, NV 89169, USA

³Institute of Engineering, Tribhuvan University, Pulchowk Campus, Kathmandu 44600, Nepal

Corresponding author: Eman Abdelfattah (abdelfattah@sacredheart.edu)

ABSTRACT Stress can disrupt daily activities and harm health if prolonged or severe. Early detection of mental stress, indicated by changes in bio-signals like thermal, electrical, and acoustic signals, can prevent related health issues. This study employs machine learning and deep learning techniques on multimodal dataset from wearable sensors, focusing on processed metrics for the three-axis acceleration (ACC), electrocardiogram (ECG), blood volume pulse (BVP), body temperature (TEMP), respiration (RESP), electromyogram (EMG), and electrodermal activity (EDA) from the 15 subjects in the WESAD dataset to effectively classify four different states – baseline, stress, amusement, and meditation. Seven traditional machine learning algorithms – Logistic Regression, Gaussian Naïve Bayes Classifier, AdaBoost Classifier, XGB Classifier, Decision Trees Classifier, Extra Trees Classifier, and Random Forest Classifier and three widely used deep learning algorithms – Deep Neural Network, Convolutional Neural Network and Recurrent Neural Network were trained and tested on the dataset on two phases to predict the state of different subject at any given time. Our findings indicate that Recurrent Neural Networks achieved an F_1 score of 93% when trained on a selected set of subjects and tested on the data from different subjects. Traditional machine learning algorithms, including Random Forest, Extra Trees, and XGB Classifiers, on the other hand, each achieved an F_1 score of 99% when trained and tested on the data for the same set of subjects. Additionally, models performed better on chest data when trained and tested on the same subjects, while they perform better on wrist data when trained on a random group of subjects and tested on the remaining ones.

INDEX TERMS Machine learning, neural networks, classification, stress detection.

I. INTRODUCTION

Stress is an inherent part of daily life, affecting individuals in various ways. While moderate stress can be managed, chronic or high levels of stress pose significant threats to safety and well-being [1]. In other words, prolonged stress can severely disrupt normal life, degrade performance in critical situations, and impair management abilities. Chronic stress is associated with various health issues, including cardiac arrhythmias, depression, and a weakened immune system, which can increase the risk of severe illnesses such as cancer [2]. In the workplace, stress is a major concern, with a significant proportion of workers reporting high stress levels and a need for better stress management strategies. An effective automated

stress detection system can provide timely interventions to mitigate these adverse effects.

Various physiological signals, such as electrodermal activity (EDA) and heart rate variability, have been shown to be reliable indicators of stress, which is manifested when an individual is overwhelmed by high demands, resulting in physical, mental, and emotional symptoms [3]. These physiological signals are highly sensitive to stress due to their strong correlation with the sympathetic nervous system and can be measured through sensors placed on different body parts, making it suitable for integration with wearable devices. These minimally intrusive devices, which are already popular among users, provide an ideal platform for continuous stress monitoring.

However, achieving high accuracy in stress detection often involves using numerous sensors and employing a vast

The associate editor coordinating the review of this manuscript and approving it for publication was Filbert Juwono⁴.

number of features, which can lead to substantial processing requirements [4]. Optimizing these systems for efficiency is not only crucial for effective prediction but also for faster and low-compute analysis. As much as the optimization of features is important, so is the choice of algorithms that suits a particular case. This research focuses on leveraging machine learning and deep learning algorithms to detect stress through physiological preprocessed data.

Our research utilizes the seven traditional machine learning models – Logistic Regression (LR), Gaussian Naïve Bayes Classifier (GaussianNB), AdaBoost Classifier (AB), XGB Classifier (XGBoost), Decision Trees Classifier (Decision Trees), Extra Trees Classifier (Extra Trees), and Random Forest Classifier (Random Forest) and three widely used deep learning algorithms – Deep Neural Network (DNN), Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) to train and test on the sensor data from the WESAD dataset, a comprehensive publicly available dataset that includes multimodal physiological signals recorded during different affective states (baseline, stressed, amusement, and meditation) [5]. Our research is conducted in two phases. Phase 1 involves training the models on a randomly selected subject and testing on the remaining set of subjects. On the other hand, phase 2 involves training and testing on combined data for all the subjects after an 80-20 train test split. Both the phases of this research are completed on the chest and wrist data and their results are analyzed to assess the efficiency of different models based on the conditions set by the phases.

A. RELATED WORK

Automated stress detection has been a long-standing pursuit, with continuous efforts aimed at refining methodologies over time. Among the most prevalent approaches is sensor-based stress detection, widely embraced for its effectiveness in capturing and analyzing stress levels. Researchers and developers have diligently sought optimal techniques to enhance accuracy and usability in this field.

Ghosh et al. proposed an image-encoding-based deep neural network model for detecting mental stress [6]. The authors conducted two experiments on both wearable stress and affect detection (WESAD) dataset and SWELL dataset. They achieved 94.8% accuracy for WESAD dataset while achieved 99.39% accuracy for SWELL dataset.

Fang et al. used electrocardiogram (ECG), electrodermal activity (EDA), and electromyogram (EMG) from WESAD dataset [7]. The authors proposed using lower overlapping windows to overcome over-fitting issue while applying machine learning models. Also, they performed normalization per subject as they did not normalize the entire dataset to boost the performance of the Random Forest (RF), Support Vector Machine (SVM), and Artificial Neural Network (ANN) classifiers. They conducted a binary classification and achieved accuracy up to 97.8%.

Hsieh et al. applied extreme gradient boosting (XGBoost) machine learning model on nine dominant features in stress detection based on Electrodermal Activity (EDA) [8].

Also, the authors performed less overlapped signal segmentation to boost the performance of XGBoost. They achieved high values of f1-scores of 92.38% and 89.92% on chest-based EDA and wrist-based EDA signal respectively.

Praveenkumar et al. used WESAD dataset to classify stress and non-stress state [9]. The authors performed two experiments: Deep Neural Network (DNN) and Long Short-Term Memory (LSTM) classifiers. In their experiments, LSTM model outperformed DNN under their utilized parameters.

Rizwan et al. extracted ECG biosignal, detected ECG Derived Respiration (EDR), derived RR interval, and QT interval to detect stress [10]. The authors used Physionet's "drivedb" dataset for training and validation for different types of Support Vector Machine (SVM) machine learning model for binary classification. They obtained 98.6% accuracy with Gaussian Kernel function.

Mentis et al. introduced an overview of some of the Artificial intelligence (AI) and Machine Learning (ML) applications for solving biomedical issues related to psychological stress and post-traumatic stress disorder (PTSD) [11]. The authors based on their study supported the use of ML for predicting stress and PTSD. They suggested the use of Swarm Intelligence (SI) for detecting stress and PTSD.

Ng et al. applied six machine learning models to predict next-day physiological and perceived stress after collecting data from sixteen pregnant women over a period of twelve weeks [12]. These participants answered 2838 ecological momentary assessment (EMA) in addition to sensor-based and intervention-based features. They concluded that random forest model outperformed the other five models. It predicted next-day physiological stress with F_1 score of 0.84 and F_1 score of 0.74 in case of next-day perceived stress.

Hartley et al. explained that in persons being treated for hypertension, whether by pharmacologic or nonpharmacologic means, caffeine may elevate blood pressure during a portion of the day, potentially reducing the effectiveness of therapy [13]. Thus, the authors suggested dietary restriction of caffeine in patients who are under evaluation for hypertension.

Schmidt et al. introduced wearable stress and affect detection (WESAD) dataset that includes recording of physiological and motion signals collected for fifteen subjects [14]. After extracting features, the authors conducted two experiments: a binary classification (stress versus non-stress) and a three-class classification (baseline, stress, and amusement). Also, they performed sixteen different modality combinations. By applying the following machine learning models: Decision Tree (DT), Random Forest (RF), AdaBoost (AB), Linear Discriminant Analysis (LDA) and k -Nearest Neighbor (k NN), they achieved in the binary classification and in the three-class classification up to 93 % and up to 80% accuracy respectively.

Mitro et al. designed, implemented, and tested an AI-Enabled Smart Wristband device capable of providing real-time stress detection during critical and emergency events [15]. The authors used WESAD dataset for training

different machine learning models for stress detection. They used the Synthetic Minority Oversampling Technique to solve the challenge of data imbalance. They achieved 96% accuracy using machine learning model/pipeline of Min-max + PCA + XGB.

Schmidt et al. reviewed the state-of-the-art wearable affect and stress recognition [16]. They aimed to encourage researchers to develop wearable systems. Schmidt et al. aimed to classify a multi-target affective state: arousal (Low, Medium, High), State-Trait Anxiety Inventory (Low, Medium, High), stress (No stress, Stressed), and valence self-reports (Negative, Neutral, Positive) [17]. The authors used data collected from eleven healthy subjects using their Ecological-Momentary-Assessments (EMAs) App during the field study that they conducted. They concluded that Convolutional Neural Networks (CNNs) outperformed the other machine learning models with low value of F_1 score of 45%. Thus, they discussed the challenges of collecting the physiology-based affective state not in lab setting.

Castro-García et al. designed a Body Area Network (BAN) with four low-cost wearables to collect physiological signals: electroencephalography (EEG), electrocardiography (ECG), breathing rate (BR), electrodermal activity (EDA), and skin temperature (ST) [18]. The reliability of BAN was demonstrated by achieving battery lifetime greater than fifteen hours, the ECG circuit has signal-noise ratio (SNR) of 9.8 dB, and the EDA has SNR of 61.6 dB.

Zanetti et al. used wireless wearable devices to extract physiological data of electrocardiogram, respiratory signal, blood volume pulse, and electroencephalogram [19]. The authors applied the random forest classifier as a supervised machine learning model. The classifier was trained to distinguish between a baseline resting condition and a mental arithmetic task and it achieved an accuracy of 97.5%.

Vavrinsky et al. presented the concept of using telemedicine device for long-term monitoring of people in a real life [20]. They suggested to measure several physiological variables including electrodermal activity, heart rate and respiration, body temperature, blood pressure using a multi-sensors device. They aimed to develop, implement, and verify their proposed concept.

Patel presented a survey of using different machine learning models to detect stress [21]. They explained that PhotoPlethysmoGram (PPG), Galvanic Skin Response (GSR), HRV, ElectroCardioGraphy (ECG) and temperature can be utilized to detect stress.

Carbonaro et al. developed a wearable biosensor platform to collect physiological and behavioral data for the treatment of stress-related disorders [22].

Kyriakou et al. proposed a rule-based algorithm based on galvanic skin response and skin temperature. The authors wanted to detect moments of stress (MOS) [23]. They concluded that their algorithm detected MOS with 84% accuracy, and it can be used to enhance stress detection in real life.

Choi et al. developed a wearable sensor platform to collect physiological features that show good correlation with mental

stress [24]. The authors applied a logistic regression model to classify the two psychophysiological conditions as either mental stress or relaxation. They achieved a success rate of 81% across subjects.

Saganowski et al. proposed a system that utilizes both smart-wearables to record physiological signals and smart-phones to gather self-assessments [25]. The authors collected the data for eight subjects within two weeks after the removal of seven subjects from the study. These seven subjects encountered technical problems that prevented them to receive the self-assessment notification to fill out the questionnaire. They concluded that their system is feasible to be used for collecting emotional data in everyday life scenarios.

Acerbi et al. analyze physiological data collected by wearable sensors [26]. They presented significant differences between stressed and not stressed people based on features extracted from heart rate variability and galvanic skin response.

Ahuja et al. experimented with four supervised machine learning models: Linear Regression, Naive Bayes, Random Forest, and Support Vector Machine and performed 10-Fold Cross Validation to predict stress [27]. They obtained the dataset from Jaypee Institute of Information Technology. They reported that Support Vector Machine classifier outperformed the other models and achieved accuracy of 85.71%, specificity 100%, and sensitivity of 75%.

Panicker et al. presented a survey with respect to important aspects of Stress Detection systems [28]. A summary of physiological data collection, Emotion Detection systems using machine learning, Stress Detection systems using machine learning, evaluation measures of these systems, the challenges in developing these systems and the variety of applications was included. Also, they proposed a multimodal system that applies machine learning models to predict stress using physiological signals.

Gedam et al. presented a review on mental stress detection using wearable sensors and machine learning [29]. Each of driving, academic, and office-like working environment was addressed and summarized from the exciting literature. Also, the authors proposed a system to detect a multimodal stress and suggested a wearable sensor-based deep learning approach to be used.

Keshan et al. performed comparative study among ten machine learning models [30]. The authors applied Naive Bayes, Logistic Regression, Multilayer Perceptron, SMO (support-vector machine), IB1 (1-nearest neighbor), IBK (K-nearest neighbors), ZeroR, J48 (decision tree), Random Forest and Random Tree on Electrocardiogram (ECG) signals to classify low stress, medium stress, or high stress. They achieved 88.24% accuracy in detecting the three classes of stress and concluded that better accuracy can be achieved in case of detecting the level of high stress.

Rescio et al. used a deep learning approach [31]. The authors implemented three neural networks on two stress categories. They reported that one dimensional convolutional

TABLE 1. Comparison of this work with other related work.

ARTICLE	ML	Extracted features	Year of Publication	Dataset / # of labels	Best F ₁ Score
<i>Ghosh et al.</i> [6]	CNN	Chest	2022	WESAD / 4 SWELL / 3	0.95 0.99
<i>Fang et al.</i> [7]	RF, SVM, ANN	Chest	2022	WESAD / 2	0.969
<i>Hsieh et al.</i> [8]	XGBoost	Chest Wrist Only Electrodermal Activity (EDA),	2019	WESAD / 2	0.9238 0.8992
<i>Praveenkumar et al.</i> [9]	LSTM, DNN	Chest	2023	WESAD / 2	Accuracy 0.86
<i>Rizwan et al.</i> [10]	SVM	ECG	2019	Physionet's "drivedb" / 2	Accuracy 0.972
<i>Schmidt et al.</i> [14]	DT, RF, kNN, Linear Discriminant Analysis (LDA), AB	Chest Wrist	2018	WESAD / 2 WESAD / 3	0.9107 0.8405 0.7549 0.6512
<i>Mitro et al.</i> [15]	XGBoost with a min-max scaler + a PCA	Chest Only PPG	2023	WESAD / 2	0.94
<i>Proposed work</i>	LR, NB, AB, XGBoost, DT, RF, ET, CNN, DNN, RNN	Chest Wrist	2024	WESAD / 4	0.998 0.998

neural network obtained 95.38% accuracy and outperformed the other neural networks.

Fontes et al. presented Hybrid Deep Learning (DL) networks for stress detection based on remote photoplethysmography [32]. The authors used for stress detection UBFC-Phys dataset that includes facial videos. Their Hybrid Deep Learning (DL) networks included Long Short-Term Memory, Gated Recurrent Units, one dimensional Convolutional Neural Network. Their approach achieved 95.83% accuracy.

Khan et al. collected data from motion sensors (accelerometers, gyroscopes, and magnetometers) on arm's wrist-worn smartphone [33]. The authors performed data preprocessing and transformation for the collected data. Then, they employed deep neural networks for activity recognition. They reported that deep neural networks achieved 77.78% real-time accuracy.

Lazarou et al. conducted a comprehensive study about the emerging field of real-time stress prediction that focused on utilizing physiological data collected by wearable devices [34]. They presented also the challenges and future directions in this domain.

The comparison of past related works and this research is shown in Table 1.

This research offers the following key contributions:

1. This research provides an extensive evaluation of both machine learning and deep learning models for

stress detection. It utilizes feature-engineered categories derived from raw sensor data in Human Activity Recognition (HAR), focusing on commonly used features such as statistical, time-domain, and frequency-domain features.

2. The study evaluates models on all four stress categories found in the WESAD dataset: baseline (0), which represents a neutral state with no stress; stress (1), which represents an induced stress state; amusement (2), which reflects a positive emotional state; and meditation (3), which represents a relaxed, calm state. Unlike many previous studies that focus solely on stress detection, this research also considers emotional states like amusement and relaxation.
3. The performance of models is assessed on data from two different sensor locations: wrist and chest. While most previous research has focused solely on chest data, this study investigates whether wrist data can also be effectively used for stress detection.
4. This research compares two data utilization approaches: training and testing on the same set of subjects, which is the traditional approach where both training and test data come from the same subjects, and training on one set of subjects and testing on a different, unseen set, which presents a more challenging scenario to test the model's generalization and adaptability to new, unseen data.

II. MACHINE LEARNING ALGORITHMS

The description of the 10 different models used in this research are provided in the subsections below:

A. MACHINE LEARNING ALGORITHMS

Logistic Regression (LR) is a linear model commonly used for binary classification tasks. It predicts the probability of occurrence of an event by fitting data to a logistic function, which maps a linear combination of input features into a range between 0 and 1 [35]. It employs a logistic (sigmoid) activation function to ensure the output probability lies within this range, making it suitable for predicting categorical outcomes based on continuous or categorical input variables.

Gaussian Naïve Bayes (NB) Classifier is a probabilistic model based on Bayes' theorem with the assumption of independence between features [36]. It is widely used for classification tasks with continuous input features, where it calculates the posterior probability of a class given the input data using the Gaussian distribution for each class's likelihood.

AdaBoost is an ensemble learning method that combines multiple weak classifiers to create a strong classifier. It sequentially trains a series of weak learners, adjusting the weights of incorrectly classified instances at each iteration to emphasize the difficult cases [37]. The final prediction is made by a weighted majority vote of the weak learners.

XGBoost (Extreme Gradient Boosting) is an optimized implementation of gradient boosting machines. It builds an ensemble of weak learners (usually decision trees) sequentially, where each subsequent learner corrects the errors of its predecessor. XGBoost employs a regularized objective function and hardware optimization to achieve high computational efficiency and predictive accuracy [38].

Decision Trees (DT) are non-linear models that recursively split the data into subsets based on the value of input features, aiming to maximize information gain or minimize impurity at each node [39]. Each internal node represents a decision based on a feature, and each leaf node represents a class label or regression value. Decision Trees can be prone to overfitting, which can be mitigated by ensemble methods like Random Forests and Gradient Boosting.

Random Forest (RF) is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes (classification) or the mean prediction (regression) of the individual trees [40]. Each tree is built using a subset of the training data and a subset of the features, introducing randomness that helps to prevent overfitting. Random Forests are versatile, easy to use, and effective for both classification and regression tasks, capable of handling large datasets with high-dimensional features.

Extra Trees (ET), or Extremely Randomized Trees, is an ensemble learning method like Random Forests but with a higher level of randomness in node splitting. Instead of searching for the best split points, Extra Trees randomly selects thresholds for each feature at each node.

This randomization reduces variance and can lead to faster training times compared to Random Forests [41]. Extra Trees are widely used for high-dimensional data.

B. DEEP LEARNING ALGORITHMS

Deep Neural Networks are multi-layered neural networks composed of interconnected nodes (neurons) organized into layers. They consist of an input layer, one or more hidden layers, and an output layer. DNNs employ non-linear activation functions such as sigmoid, tanh, or ReLU to capture complex relationships in data and are trained using backpropagation to minimize prediction error [42]. DNNs excel in learning hierarchical representations of data, making them suitable for tasks like image and speech recognition, natural language processing, and reinforcement learning.

Convolutional Neural Networks are specialized deep learning models designed for processing grid-like data, such as images or audio spectrograms. CNNs consist of convolutional layers that apply filters (kernels) across input data to extract spatial hierarchies of features, followed by pooling layers to downsample and reduce spatial dimensions [43]. They leverage weight sharing and local connectivity to efficiently learn spatial hierarchies of features, making them highly effective in tasks like image classification, object detection, and image segmentation.

Recurrent Neural Networks are designed to process sequential data by maintaining an internal state (memory) to capture dependencies between elements in the sequence. RNNs have feedback connections that allow information to persist, enabling them to model temporal dynamics in data [44]. They are structured with recurrent layers and use activation functions like tanh or LSTM (Long Short-Term Memory) cells to manage the flow of information over time [45]. RNNs are well-suited for tasks such as natural language processing, time series prediction, and speech recognition where sequential dependencies are crucial.

III. PERFORMANCE METRICS

Accuracy, while straightforward, can be misleading when class distributions are uneven, such as in distinguishing different states of emotions. Even if a model simply predicts the majority class, it might still show high accuracy. Hence, accuracy isn't suitable for this study's performance metric.

Precision, calculated as $\text{True Positives} / (\text{True Positives} + \text{False Positives})$, measures the proportion of correctly predicted positives among all predicted positives. Recall, given by $\text{True Positives} / (\text{True Positives} + \text{False Negatives})$, assesses the proportion of correctly predicted positives out of all actual positives. Choosing precision emphasizes minimizing false positives, while recall focuses on minimizing false negatives.

For a comprehensive evaluation of machine learning algorithms in detection of stress along with other human states, both precision and recall are crucial. Therefore, the F1 score,

which balances precision and recall using $F1 \text{ score} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$, serves as the primary performance metric in both the phases of this research.

In addition, the time taken to train and test each of the learning algorithms are also used as a measure of efficiency as model should not only be predicting the right state accurately but also must do so in a shorter amount of time to consider the large volume of sensor data that needs to be processed every second.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The WESAD dataset contained raw sensor data from two devices: RespiBAN, a chest-worn device, and Empatica E4, a wrist-worn device, collected from 15 subjects [5]. The RespiBAN signals were sampled at 700 Hz and included measurements of ECG (mV), EDA (μS), EMG (mV), TEMP ($^{\circ}\text{C}$), ACCEL for the three axes (g), and Respiration (%). Data from the Empatica E4 wrist-worn device included ACC sampled at 32 Hz (in units of 1/64g), BVP sampled at 64 Hz (PPG data), EDA sampled at 4 Hz (in μS), and TEMP sampled at 4 Hz (in $^{\circ}\text{C}$).

The sampled data were preprocessed to generate different set of features from each signal based on their effectiveness in previous research works before feeding them to the machine and deep learning algorithms. The features that were used in this research are as shown in Table 2.

The rationale behind choosing these features is as follows:

The mean and standard deviation for ACC captures overall body movement; stress increases physical restlessness, leading to increased activity. While absolute integration for ACC reflects total movement over time; stress leads to increased cumulative movement and peak frequency indicates dominant movement patterns and repetitive or abrupt motions.

Higher Heart Rate and high variability indicate stress reactivity for ECG and BVP. Also, lower HRV is associated with stress, reflecting reduced parasympathetic activity and lower values for triangular interpolation of HRV indicate stress. Furthermore, root mean square quantifies short-term HRV and higher LF/HF ratio for ECG and BVP is commonly associated with stress.

For EDA, mean and standard deviation are extracted because higher skin conductance associated with increased stress. Also, minimum and maximum values for EDA indicates reactivity to stress. Moreover, steeper slopes and wide dynamic range in EDA over time are associated with stress.

For EMG, mean and standard deviation indicates muscle activity. Also, dynamic range in EMG reflects muscle activity intensity. Furthermore, peak frequency and number of peaks captures muscle activity intensity.

For RESP, mean and standard deviation are extracted as average temperature changes with stress. Also, minimum and maximum values reflect temperature reactivity to stress. Furthermore, dynamic range and slope in RESP indicates temperature fluctuation and trend over time with stress.

TABLE 2. Preprocessed features from signal data.

Signal	Features Extracted
ACC	Mean and STD for each axis individually and combined. Absolute integrate for each axes individually and combined. Peak frequency for each axes.
ECG and BVP	Mean and STD of the HR Mean and STD of the HRV Triangular interpolation index Root mean square of the HRV Ratio of LF and HF component Relative power of frequent component Normalized LF and HF component
EDA	Mean and STD Min and max value Slope and dynamic range
EMG	Mean and STD Dynamic range Absolute integral Mean, Median, 10 th percentile, 90 th percentile, peak frequency and number of peaks Mean and STD of peak amplitudes
RESP	Mean and STD of inhalation Stretch, Volume Breath rate Respiration Duration
TEMP	Mean and STD Min and max Dynamic Range Slope

A. PHASE 1

For phase 1, ten different subjects were arbitrarily chosen, and their preprocessed features were used to train the algorithms.

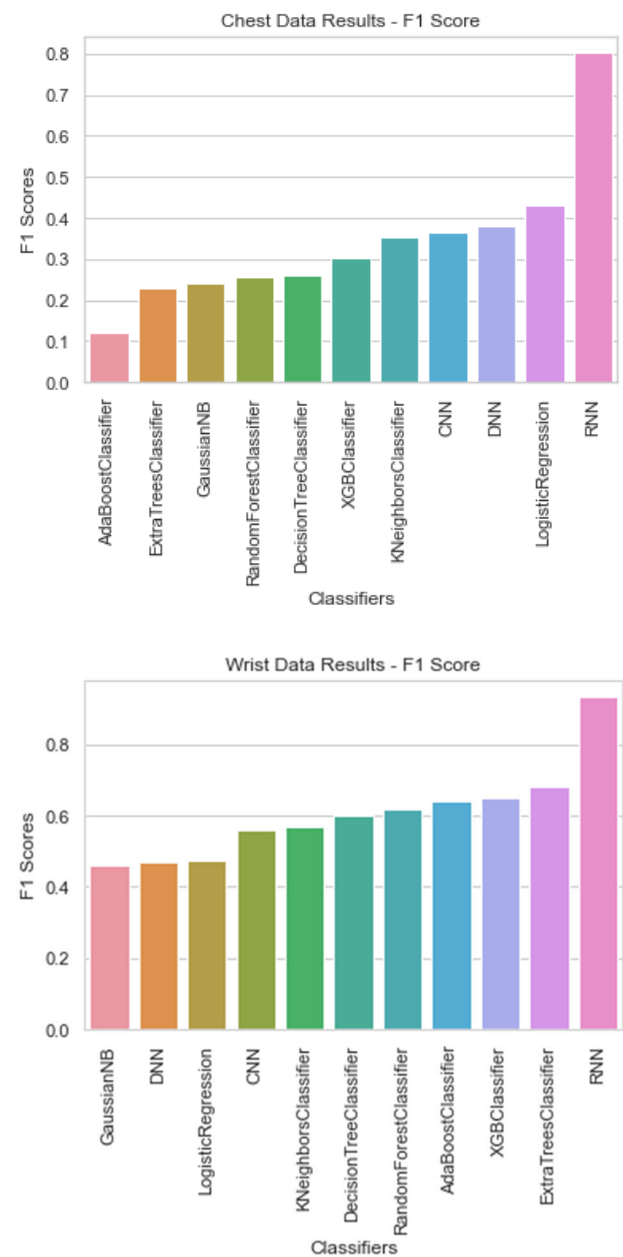


FIGURE 1. F₁ scores of the classifiers for Chest and Wrist data for Phase 1.

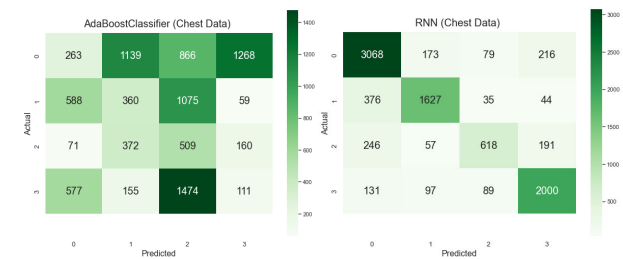


FIGURE 2. Confusion matrices of the worst and best performing classifiers on Chest data for Phase 1.

A 10-fold cross validation was employed to ensure robust model evaluations and mitigating any potential overfitting.

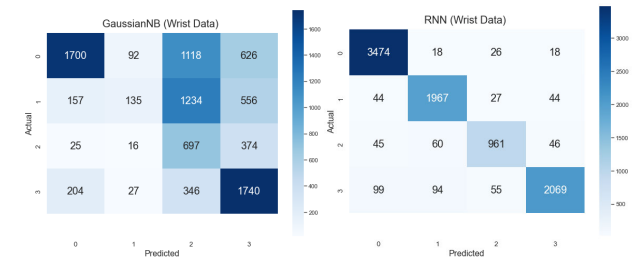


FIGURE 3. Confusion matrices of the worst and best performing classifiers on Wrist data for Phase 1.

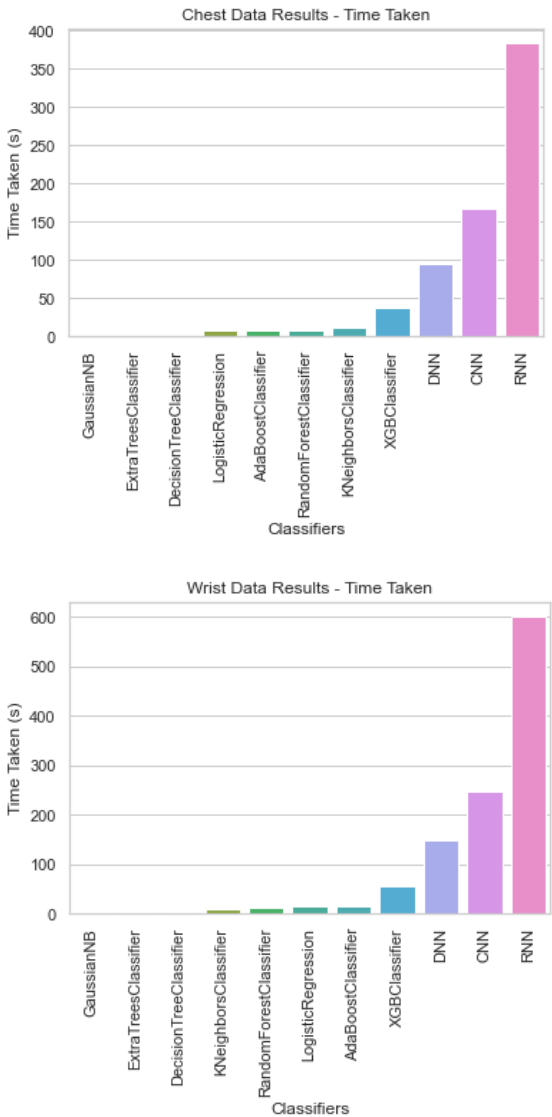


FIGURE 4. Time taken by the classifiers for Chest and Wrist data.

The train and test were executed on chest data and then on the wrist data separately.

The F₁ scores of all the classifiers for the chest and wrist data are displayed in Fig. 1. While RNN had the highest F₁ scores in both the chest and wrist data (80.8% and 93.6% respectively), Gaussian NB and AdaBoost had the lowest F₁ scores of 24.0% and 46.0% in chest and wrist

TABLE 3. Evaluation metrics of classification algorithms for phase I.

CHANNEL	CLASSIFIER/METRIC	PRECISION	RECALL	F ₁ SCORE	TIME (IN SECONDS)
Chest	Logistic Regression	49.7	46.3	43.0	6.8
	Gaussian NB	46.6	31.4	24.0	0.06
	AdaBoost	14.3	13.7	12.0	7.2
	XGBoost	38.3	30.2	30.3	37.1
	Decision Trees	30.1	28.0	25.9	1.9
	Random Forest	31.4	24.8	25.5	7.7
	Extra Trees	28.1	22.3	23.1	1.73
	DNN	43.1	35.6	38.0	94.0
	CNN	41.0	37.3	36.5	167.3
Wrist	RNN	80.8	80.8	80.8	383.8
	Logistic Regression	50.5	49.0	47.3	13.8
	Gaussian NB	59.4	47.2	46.0	0.09
	AdaBoost	61.9	67.6	64.1	14.9
	XGBoost	65.1	70.1	64.9	55.2
	Decision Trees	63.0	63.1	60.0	3.2
	Random Forest	63.2	66.7	61.9	10.9
	Extra Trees	66.5	72.8	68.0	1.99
	DNN	45.9	48.7	47.0	149.3
	CNN	56.8	57.2	56.1	247.1
	RNN	93.6	93.6	93.6	600.1

data respectively. The confusion matrices of the best and worst classifiers for chest and wrist data have been displayed in Fig. 2 and Fig. 3 respectively. It is evident from Fig. 1, Fig. 2 and Fig. 3 that RNN performed better than the traditional machine learning models and other neural networks in terms of F₁ scores.

The time that the neural networks took to train, and test were considerably higher than the same taken by the traditional machine learning models. As evident from Fig. 4, RNN took the longest time of 384 and 600 seconds while Gaussian NB took the shortest time of 0.06 and 0.09 seconds in the chest and wrist data respectively. The precision, recall, F₁ score, and time taken by classifiers for this phase of the research has been shown in Table 3.

Table 3 compares the performance and elapsed time of various algorithms in accomplishing the task. In terms of F₁ score, the Recurrent Neural Network (RNN) emerges as the clear winner for Phase I, which is expected given its ability to capture time-dependent relationships. For cross-subject analysis, RNNs are particularly effective, as they can leverage sequential data to enhance predictive performance. This advantage is evident across both chest and wrist data, where RNNs outperform other algorithms in terms of accuracy.

However, it is important to note that RNNs, like other neural network-based models, tend to be more resource- and time-intensive. Consequently, while this approach may be well-suited for applications such as long-term mental health monitoring—where Heart Rate Variability (HRV) and Heart Rate (HR) data are analyzed over time to assess patient progress—it may not be optimal for real-time stress evaluation. In contexts such as wearable devices or

emergency situations, where rapid and continuous evaluation is critical, the computational demands of RNNs may hinder their practicality.

B. PHASE 2

For phase 2, the preprocessed data from chest and wrist for all the subjects were combined and a train test split was made with the ratio 80:20. As was done in phase 1, a 10-fold cross-validation was implemented to provide more reliable model evaluations while minimizing the risk of overfitting. The train and test were executed on chest data and then on the wrist data separately.

The F₁ scores of all the classifiers for the chest and wrist data are displayed in Fig. 5. While XGBoost Classifier had the highest F₁ scores of 99.8% in both chest and wrist data, Gaussian NB had the lowest F₁ scores of 39.1% and 50.1% respectively. The confusion matrices of the best and worst classifiers for chest and wrist data have been displayed in Fig. 6 and Fig. 7 respectively.

In phase 1 of this research, the time that the neural networks took to train, and test were considerably higher than the time taken by the traditional machine learning models. As evident from Fig. 8, RNN took the longest time of 564 and 614 seconds while Gaussian NB took the shortest time of 0.13 and 0.24 seconds in the chest and wrist data respectively. The precision, recall, F₁ score and time taken by classifiers for this phase of the research has been shown in Table 4.

As evident in Table 4, XGBoost demonstrates superior performance in both F₁ score, and computational time compared to other algorithms. While not as fast as simpler models, such as decision trees, XGBoost proves to be an effective choice

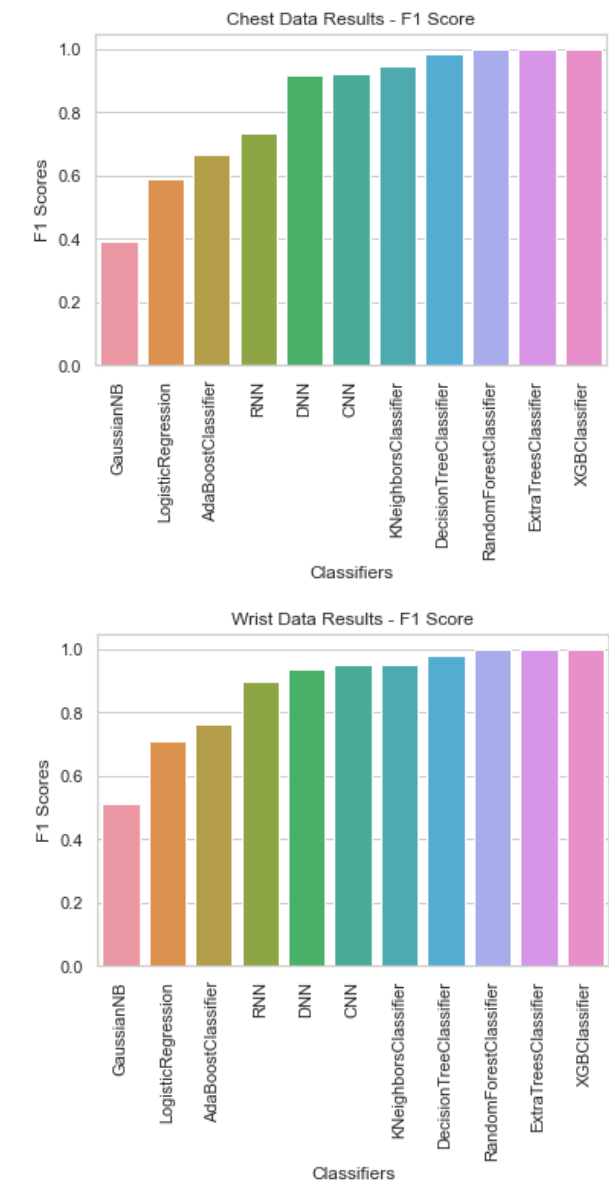


FIGURE 5. F₁ scores of the classifiers for Chest and Wrist data for Phase 2.

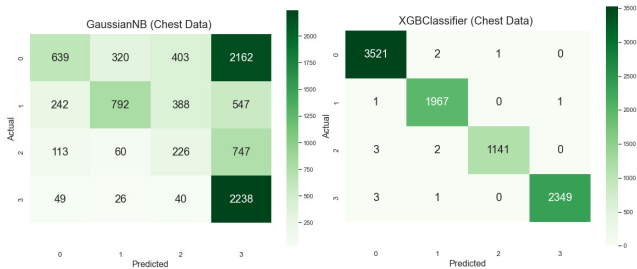


FIGURE 6. Confusion matrices of the worst and best performing classifiers on Chest data for Phase 2.

for problems where subject-specific accuracy is paramount, and time constraints are somewhat flexible. A remarkably high F₁-score of 99.8% for both chest and wrist data in phase 2 (in stark contrast to the results in phase 1) does

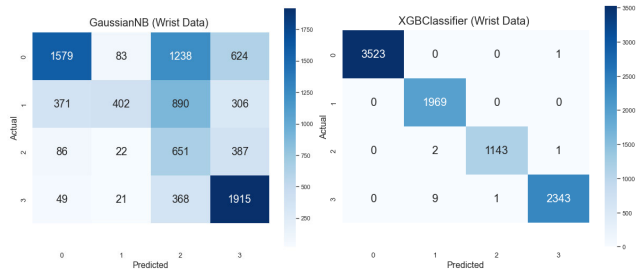


FIGURE 7. Confusion matrices of the worst and best performing classifiers on Wrist data for Phase 2.

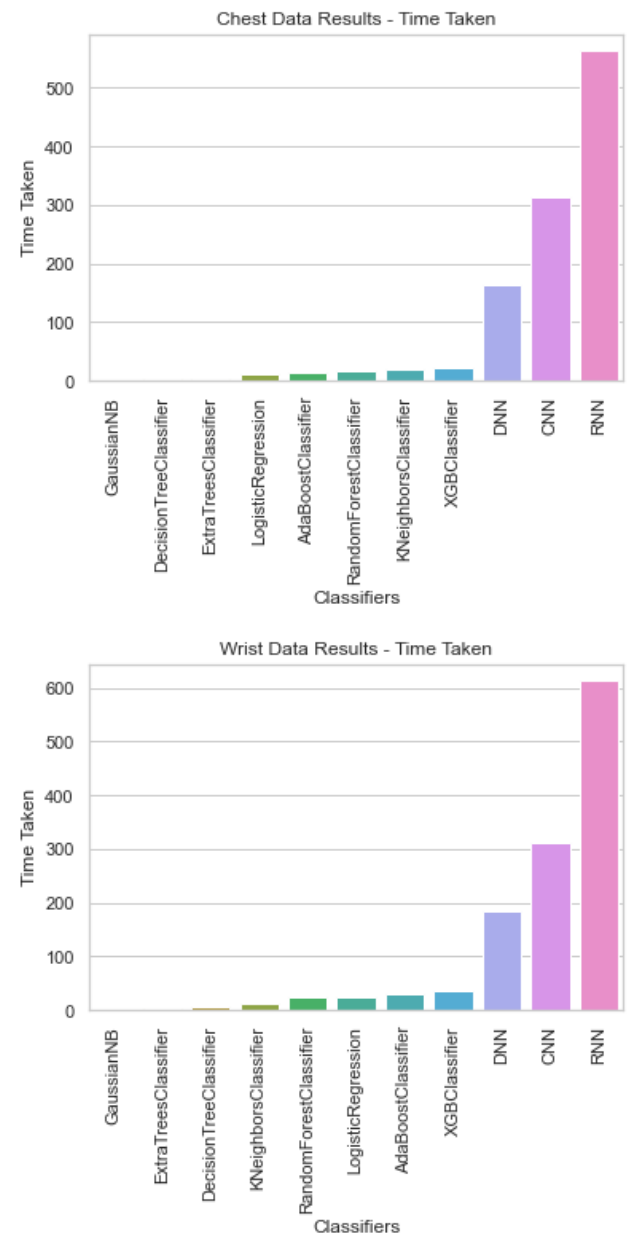


FIGURE 8. Time taken by the classifiers for Chest and Wrist data for Phase 2.

raise concerns about generalizability of the machine learning models for cross-subject applications.

TABLE 4. Evaluation metrics of classification algorithms for phase II.

CHANNEL	CLASSIFIER/METRIC	PRECISION	RECALL	F ₁ SCORE	TIME (IN SECONDS)
Chest	Logistic Regression	60.9	61.7	58.9	11.0
	Gaussian NB	51.5	43.3	39.1	0.13
	AdaBoost	66.8	67.1	66.6	14.2
	XGBoost	99.8	99.8	99.8	21.7
	Decision Trees	98.3	98.3	98.3	2.1
	Random Forest	99.6	99.6	99.6	16.7
	Extra Trees	99.7	99.7	99.7	3.0
	DNN	92.0	92.0	92.0	163.6
	CNN	92.5	92.3	92.3	312.0
Wrist	RNN	74.4	74.2	73.2	563.7
	Logistic Regression	70.7	73.1	70.1	25.0
	Gaussian NB	64.5	50.6	50.9	0.24
	AdaBoost	76.4	77.1	76.4	29.4
	XGBoost	99.8	99.8	99.8	37.2
	Decision Trees	97.9	97.9	97.9	7.5
	Random Forest	99.7	99.7	99.7	23.5
	Extra Trees	99.8	99.8	99.8	3.9
	DNN	94.0	93.7	93.8	183.4
	CNN	95.0	95.0	95.0	312.5
	RNN	90.0	90.0	90.0	614.0

V. CONCLUSION

This research leveraged the WESAD dataset, encompassing raw sensor data from RespiBAN and Empatica E4 devices, to evaluate the performance of various machine learning and deep learning algorithms in stress detection. Through a two-phase experimental setup, the preprocessed features derived from the sensor data were used to train and test several classifiers on chest and wrist data separately.

Phase 1 focused on ten arbitrarily chosen subjects. The results indicated that the RNN outperformed other models, achieving the highest F₁ scores for both chest and wrist data. Despite the superior accuracy of the recurrent neural network, its training and testing time was significantly longer than those of traditional machine learning models.

Phase 2 expanded the dataset to include training not just on a subset of subjects but all the subjects and applied an 80:20 train-test split. In this phase, XGBoost classifier achieved the highest F₁ scores for both chest and wrist data, while Gaussian NB remained the least effective. Similar to phase 1, neural networks, particularly the RNN, required considerably more time to train and test compared to traditional models.

In phase 1, the RNN outperformed other models in predicting stress categories for previously unseen subjects. In contrast, traditional machine learning models, specifically the XGBoost classifier, demonstrated superior performance when trained and tested on the same subject set, as observed in phase 2.

RNN's superior performance in phase 1 can be attributed to its ability to memorize temporal dependencies (relationships), which enables it to perform better on unseen data during cross-subject scenarios (testing). The high F₁ score achieved by XGBoost in phase 2 suggests that the model might have learned patterns specific to individuals, that may

not generalize well across different individuals. This could indicate a tendency toward overfitting when trained on consistent intra-subject data.

Thus, applications needing cross-subject generalization (e.g., wearable devices applicable for diverse individuals) are benefited a great deal by RNN's sequential modeling, which is beneficial, especially when data reflects variable physiological responses across individuals.

For individualized or controlled applications (e.g., clinical settings with repeated measures on the same subject), XGBoost is advantageous in handling structured data in terms of speed and performance.

Future research will aim to focus on common techniques such as regularization, making ensembles and hyperparameter tuning to increase the generalizability of machine learning models and performing optimization techniques, such as early stopping, adding learning rate schedulers and distributed training, on deep learning models to enhance their efficiencies in terms of training and testing time.

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EMAN ABDELFAH received the M.S. degree in computer science and the Ph.D. degree in computer science and engineering from the University of Bridgeport, in 2002 and 2011, respectively.

She was an Assistant Professor of computer science with Ramapo College of New Jersey, USA. She was a Professional Assistant Professor of computer science with the School of Engineering and Computing Sciences, Texas A&M University, Corpus Christi, TX, USA. She was also an Adjunct

Professor with the Department of Computer Science and Engineering and the Department of Mathematics, University of Bridgeport, USA. She is currently an Assistant Professor with Sacred Heart University, USA, and an Adjunct Assistant Professor with American Intercontinental University. Her research results were published in prestigious international conferences. Her research interests include machine learning, network security, and networking.



SHREEHAR JOSHI received the Bachelor of Science degree in computer science and data science from Ramapo College of New Jersey. He is currently a Software Engineer with The Boring Company. He has made significant contributions to the fields of machine learning and deep learning, publishing numerous research papers in esteemed national and international conferences. His research spans various applications of machine learning, including human activity recognition, the

IoT BotNet detection, diagnosis of medical conditions, and textual analysis.

Additionally, he has leveraged his machine learning expertise in his previous role at AUXPAD.



SHREEKAR TIWARI received the Bachelor of Engineering degree in electronics and communication engineering from Thapathali Engineering Campus. He is currently pursuing the master's degree in information and communication engineering with Pulchowk Campus. He has extensive experience working with reputable ISPs in Nepal, including Worldlink Communications and Divine Stream Networks. His professional journey has been marked by significant contributions to the

field of machine learning. He is also engaged in research and projects focusing on human activity recognition using machine and deep learning techniques, aiming to enhance the accuracy and efficiency of activity detection systems. His work in this domain involves designing algorithms, processing data, and applying advanced neural network architectures to address complex challenges in human activity monitoring.

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